

Statistical Methods for Data Science (Fall 2025)

Mini Project 3

Instructions:

- Due date: November 14, 2025.
- Total points = 20.
- Submit a typed report.
- Justify all steps and provide all relevant explanations.
- You can work on the project either individually or in a group of no more than two students. In case of the latter, submit only one report for the group, and include a description of the contribution of each member.
- It is OK to discuss the project with other students in the class (even those who are not in your group), but each group must write its own code and answers. If the submitted report (including code and answer) is similar (either partially or fully) to someone else's, this will be considered evidence of academic dishonesty, and you will be referred to appropriate university authorities.
- You must use the following template for your report:

Mini Project #

Name

Names of group members (if applicable)

Contribution of each group member

Section 1. Answers to the specific questions asked

Section 2: R code. Your code must be annotated. No points may be given if a brief look at the code does not tell us what it is doing.

1. Suppose the lifetime, in years, of an electronic component can be modeled by a continuous random variable with probability density function

$$f(x) = \begin{cases} \frac{\theta}{x^{\theta+1}} & x \geq 1, \\ 0, & x < 1, \end{cases}$$

where $\theta > 0$ is an unknown parameter. Let $X_1, \dots, X_n$ be a random sample of size n from this population.

- (a) Derive an expression for maximum likelihood estimator of θ .
- (b) Suppose $n = 5$ and the sample values are $x_1 = 21.72, x_2 = 14.65, x_3 = 50.42, x_4 = 28.78, x_5 = 11.23$. Use the expression in (a) to provide the maximum likelihood estimate for θ based on these data.
- (c) Even though we know the maximum likelihood estimate from (b), use the data in (b) to obtain the estimate by *numerically* maximizing the log-likelihood function using `optim` function in R. Do your answers match?
- (d) Use the output of numerical maximization in (c) to provide approximate standard error of the maximum likelihood estimate and an approximate 95% confidence interval for θ . Are these approximations going to be good? Justify your answer.